Student Mental Health & Burnout

Machine Learning Analysis Report

Dataset	student_mental_health_burnout.csv · 150,000 records
Target Variable	burnout_level (High / Medium / Low) — Multi-class Classification
Features (X)	age, CGPA, sleep hours, study hours, screen time, anxiety score, depression score, academic pressure, financial stress, social support, physical activity, attendance, gender, course, year, stress level, sleep quality, internet quality
Models	Decision Tree · Naïve Bayes · K-Nearest Neighbors (k=5) · Logistic Regression
Train / Test	80% training / 20% testing · stratified split · random_state = 42
Scaling	StandardScaler applied for KNN and Logistic Regression

Libraries: pandas · numpy · matplotlib · seaborn · scikit-learn

1. Dataset Overview

The dataset used in this analysis is the Student Mental Health & Burnout dataset sourced from Mansehaj Preet on Kaggle. It contains 150,000 student records across 20 columns, covering a mix of demographic, academic, and mental health variables. There are no missing values, and each row represents a unique student.

Numerical features (12): age, daily study hours, daily sleep hours, screen time hours, anxiety score, depression score, academic pressure score, financial stress score, social support score, physical activity hours, attendance percentage, and CGPA.

Categorical features (6): gender, course, year of study, stress level, sleep quality, and internet quality.

The dataset is well-suited for classification because it combines both types of variables, has a sufficiently large sample size (150,000 records), and contains a clearly defined target variable with three classes that reflect real-world burnout levels.

Kaggle Link: http://bit.ly/4bKuJuX

2. Determining X (Features) and Y (Target)

Y — Target Variable:

burnout_level is the column the models are trained to predict. It has three possible values: Low, Medium, and High, making this a multi-class classification problem. The target is Label-Encoded to integers (High = 0, Low = 1, Medium = 2) so scikit-learn can process it.

X — Feature Variables:

All remaining columns, after removing student_id (a row identifier with no predictive value), are used as input features. This gives 18 features per student. A combination of raw numerical measurements and encoded categorical labels.

Encoding & Scaling:

Categorical columns (gender, course, year, stress_level, sleep_quality, internet_quality) are converted to integers via Label Encoding. For Decision Tree and Naïve Bayes, features are used as-is. These algorithms are not affected by feature scale. For KNN and Logistic Regression, all features are standardised using StandardScaler (mean = 0, standard deviation = 1) to ensure that features with larger numerical ranges do not dominate the model.

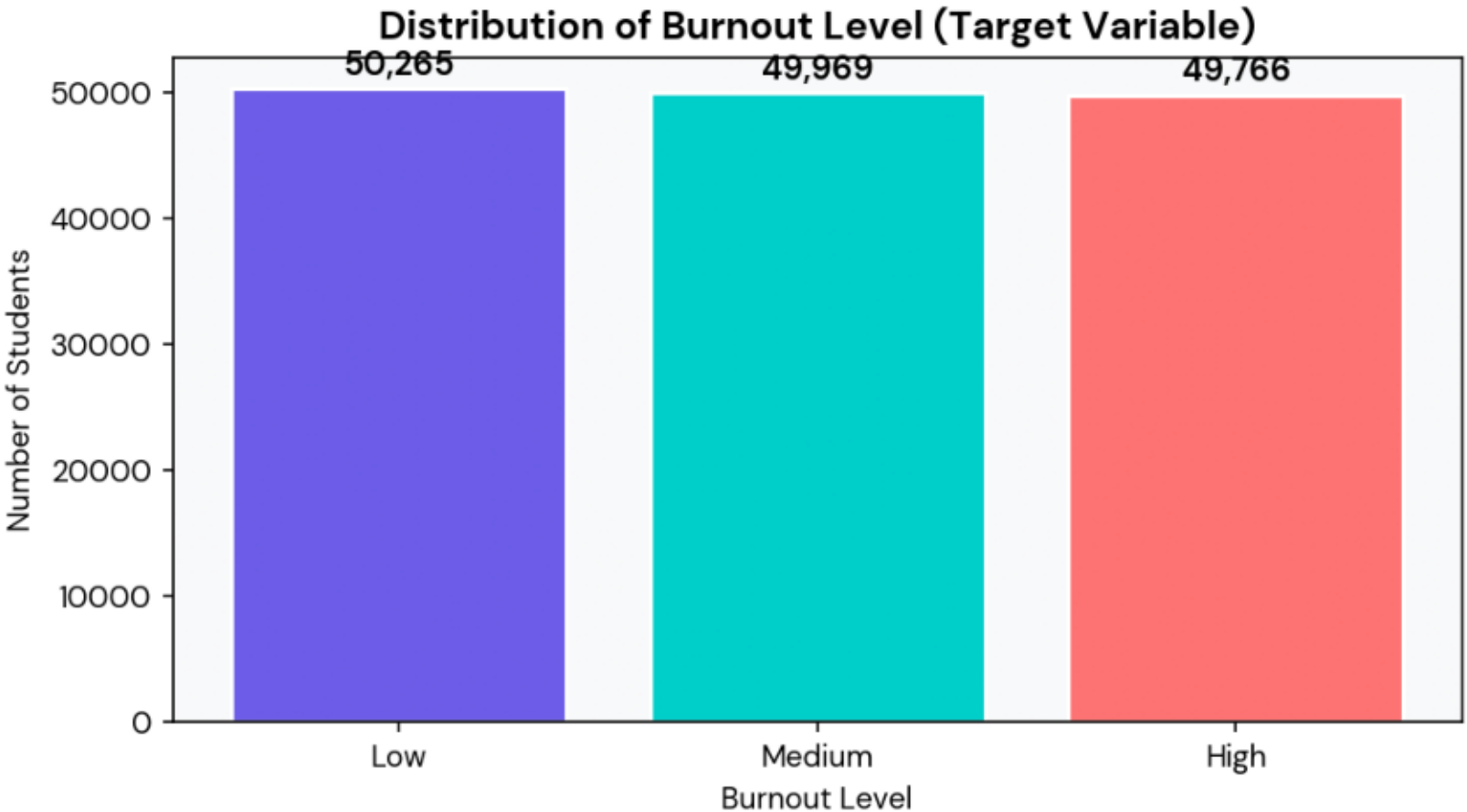

Chart 1 – Burnout Level Distribution

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This bar chart shows the number of students in each burnout category: Low, Medium, and High. Checking the class balance is the first step before training any classifier — a heavily skewed distribution can cause models to favour the majority class. A roughly equal distribution means the models can learn meaningful patterns for all three burnout levels.

```
counts = df["burnout_level"].value_counts().reindex(order)
bars = ax.bar(order, counts.values, color=palette, edgecolor="white", linewidth=1.5)
for bar, val in zip(bars, counts.values):
    ax.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 500,
            f"{val:,}", ha="center", va="bottom", fontsize=11, fontweight="bold")
ax.set_title("Distribution of Burnout Level (Target Variable)")
```

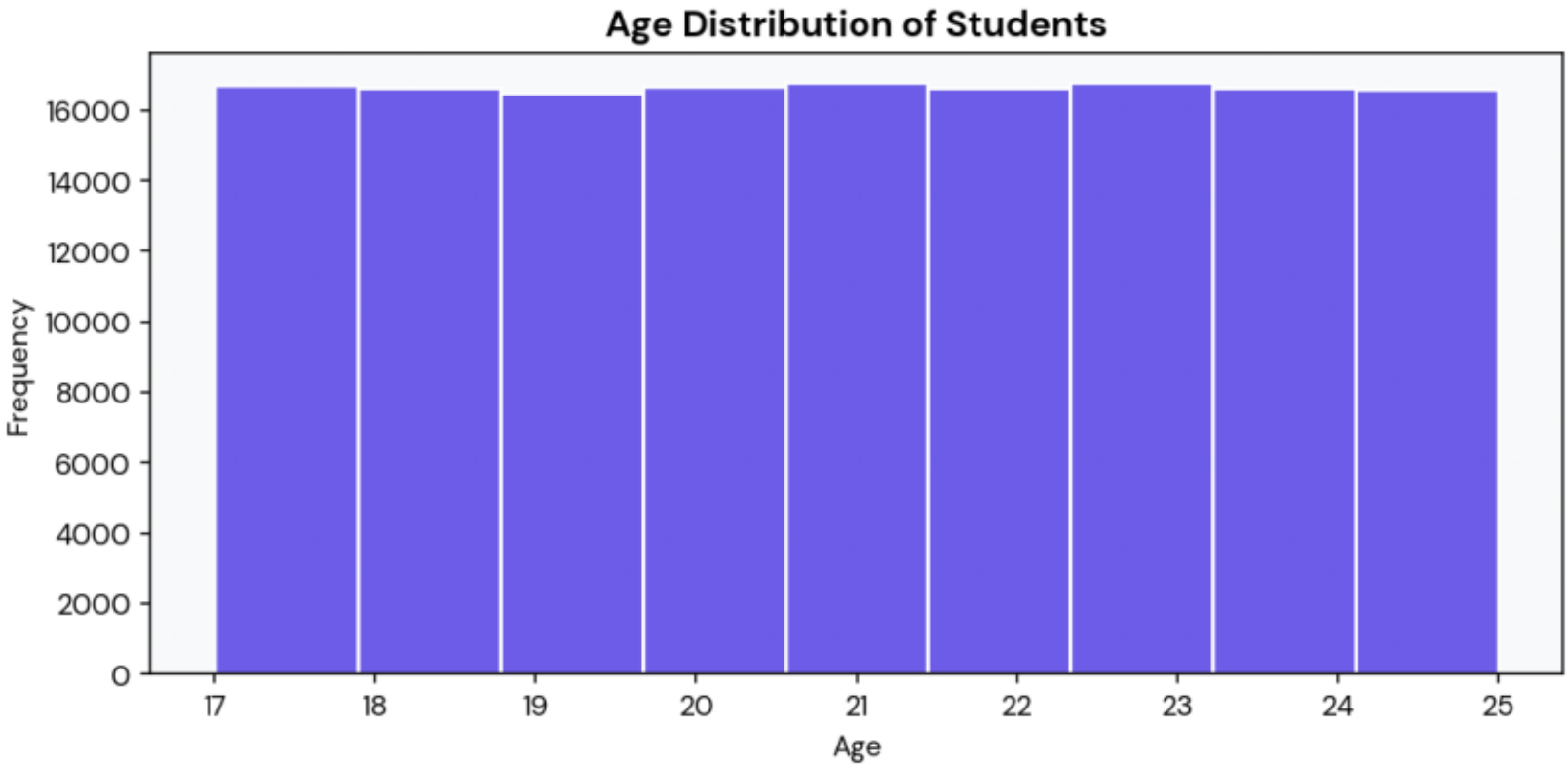

Chart 2 – Age Distribution

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This histogram shows the age range of students in the dataset (17–25 years). The narrow range is expected for a university-level study population. Age is kept as a feature because coping strategies and academic pressure may differ as students progress through their studies, which can influence burnout outcomes.

```
ax.hist(df["age"], bins=9, color="#6C5CE7", edgecolor="white", linewidth=1.3)
ax.set_title("Age Distribution of Students")
ax.set_xlabel("Age")
ax.set_ylabel("Frequency")
```

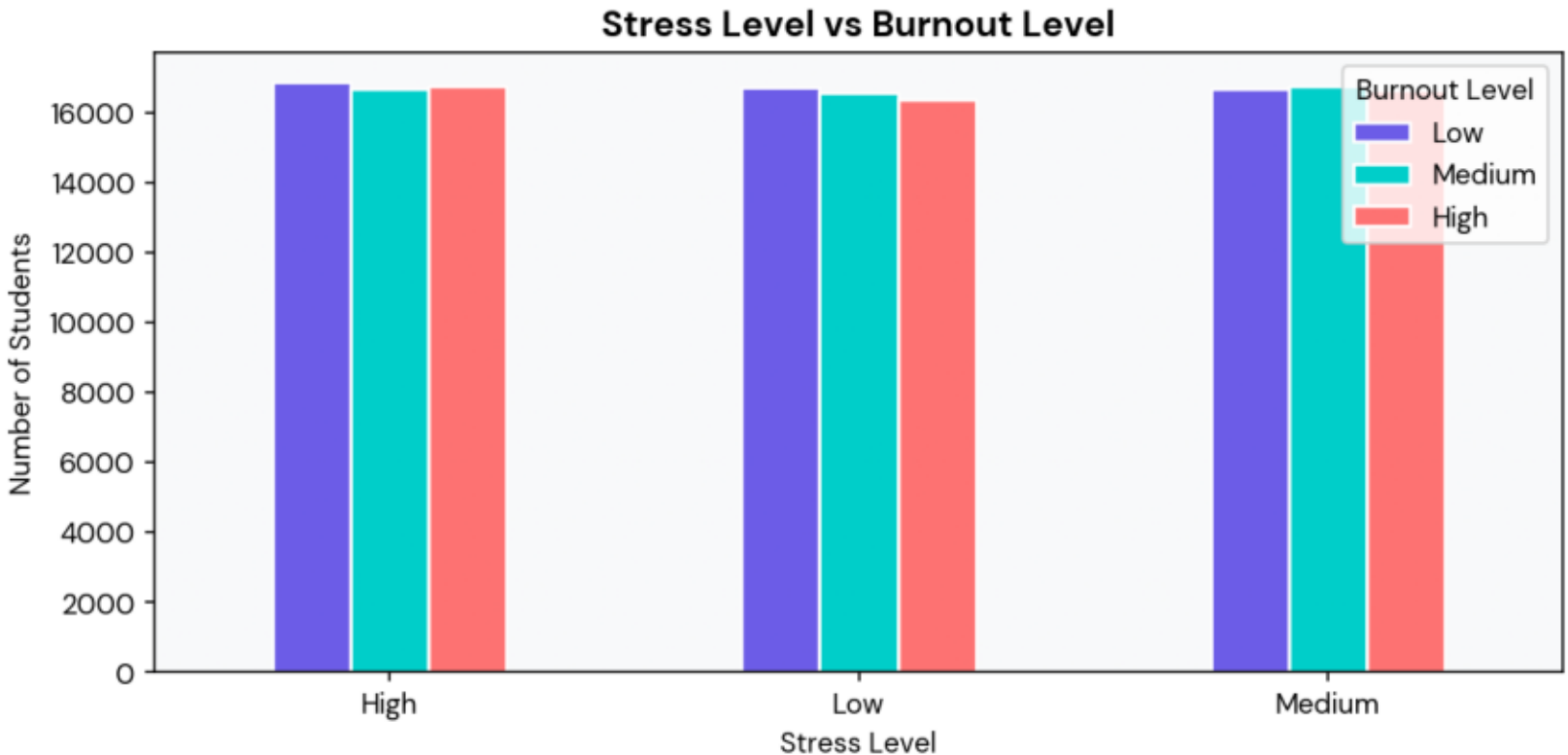

Chart 3 – Stress Level vs Burnout Level

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This grouped bar chart crosses two categorical variables: self-reported stress level (Low / Medium / High) and burnout level. It visually tests whether students who report high stress are disproportionately represented in the High burnout group. A strong visible pattern confirms that stress level is a meaningful predictor for burnout.

```
ct = pd.crosstab(df["stress_level"], df["burnout_level"])[order]
ct.plot(kind="bar", ax=ax, color=palette, edgecolor="white", linewidth=1.2)
ax.set_title("Stress Level vs Burnout Level")
ax.legend(title="Burnout Level")
```

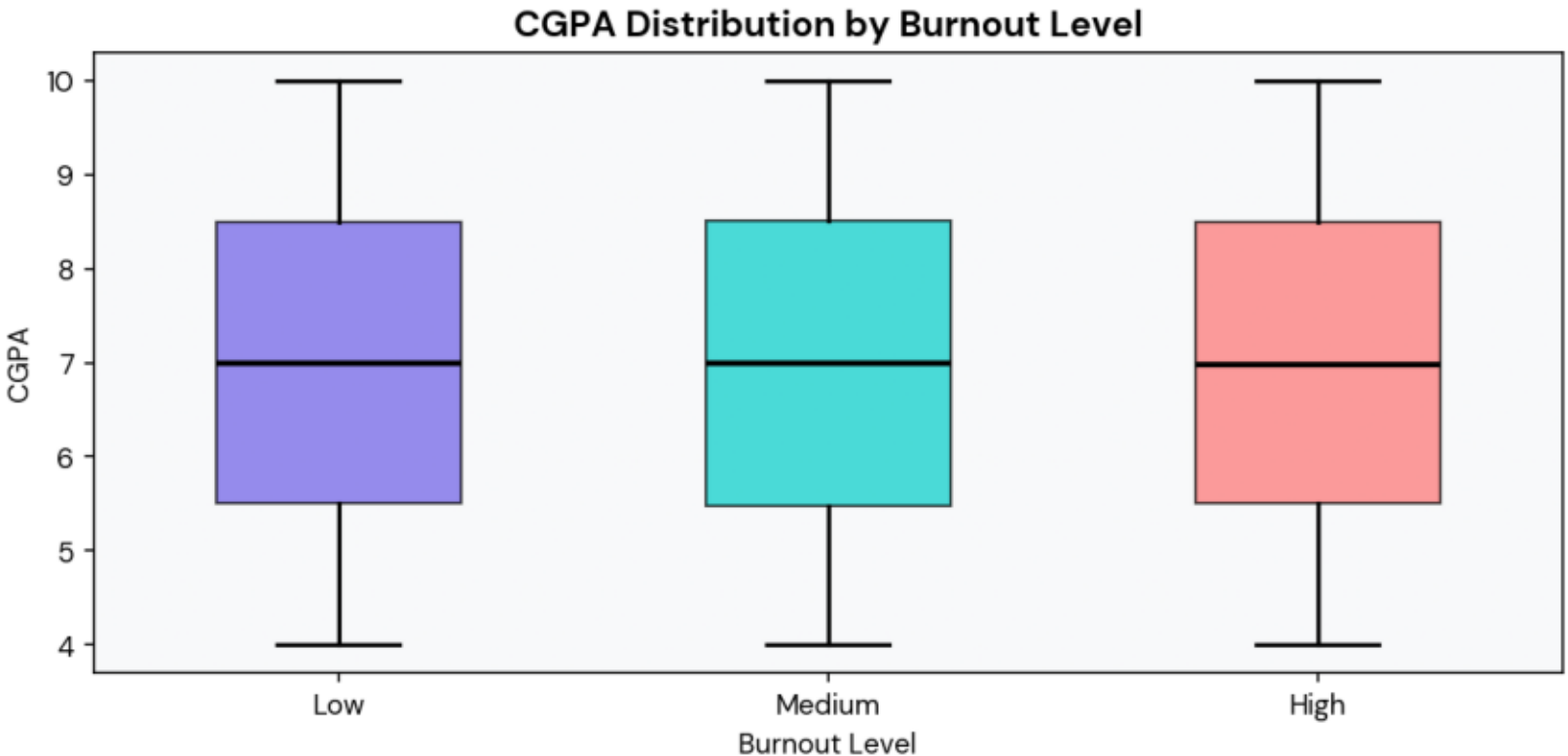

Chart 4 – CGPA by Burnout Level

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These box plots compare the distribution of CGPA (Cumulative Grade Point Average) across the three burnout groups. The box spans the 25th to 75th percentile, the line inside is the median, and the whiskers cover the data range. Differences in median CGPA between groups indicate that academic performance is a useful predictor.

```
for i, (level, color) in enumerate(zip(order, palette)):
    data = df[df["burnout_level"] == level]["cgpa"]
    ax.boxplot(data, positions=[i], patch_artist=True, widths=0.5)
ax.set_title("CGPA Distribution by Burnout Level")
```

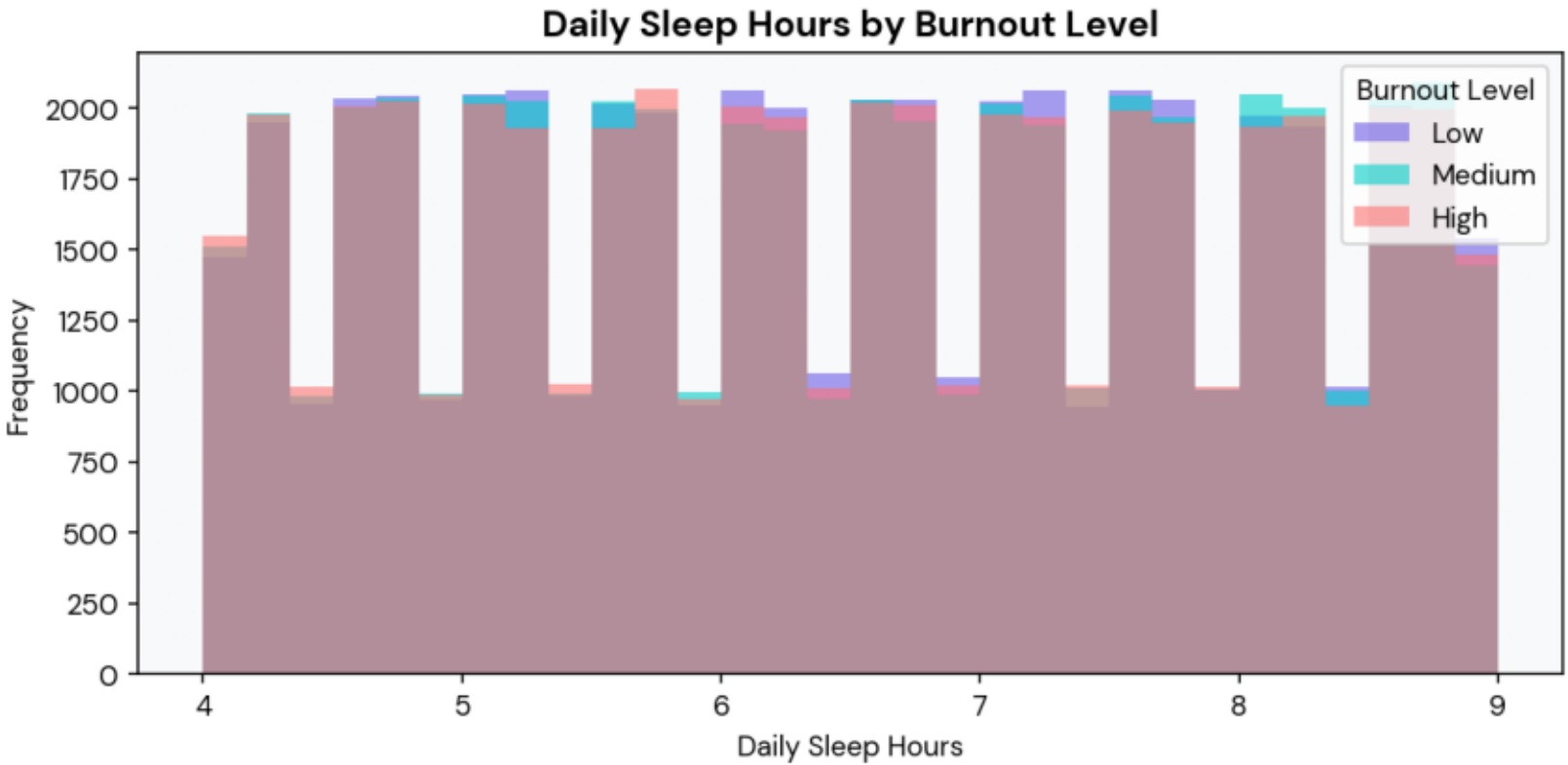

Chart 5 – Daily Sleep Hours by Burnout Level

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These overlapping histograms compare the number of hours per day students sleep, grouped by burnout level. If students with High burnout consistently sleep fewer hours, sleep duration is a strong predictor. The overlap between distributions reflects real-world variability and is well handled by both probabilistic (Naïve Bayes) and distance-based (KNN) classifiers.

```
for level, color in zip(order, palette):
    data = df[df["burnout_level"] == level]["daily_sleep_hours"]
    ax.hist(data, bins=30, alpha=0.6, label=level, color=color)
ax.set_title("Daily Sleep Hours by Burnout Level")
ax.legend(title="Burnout Level")
```

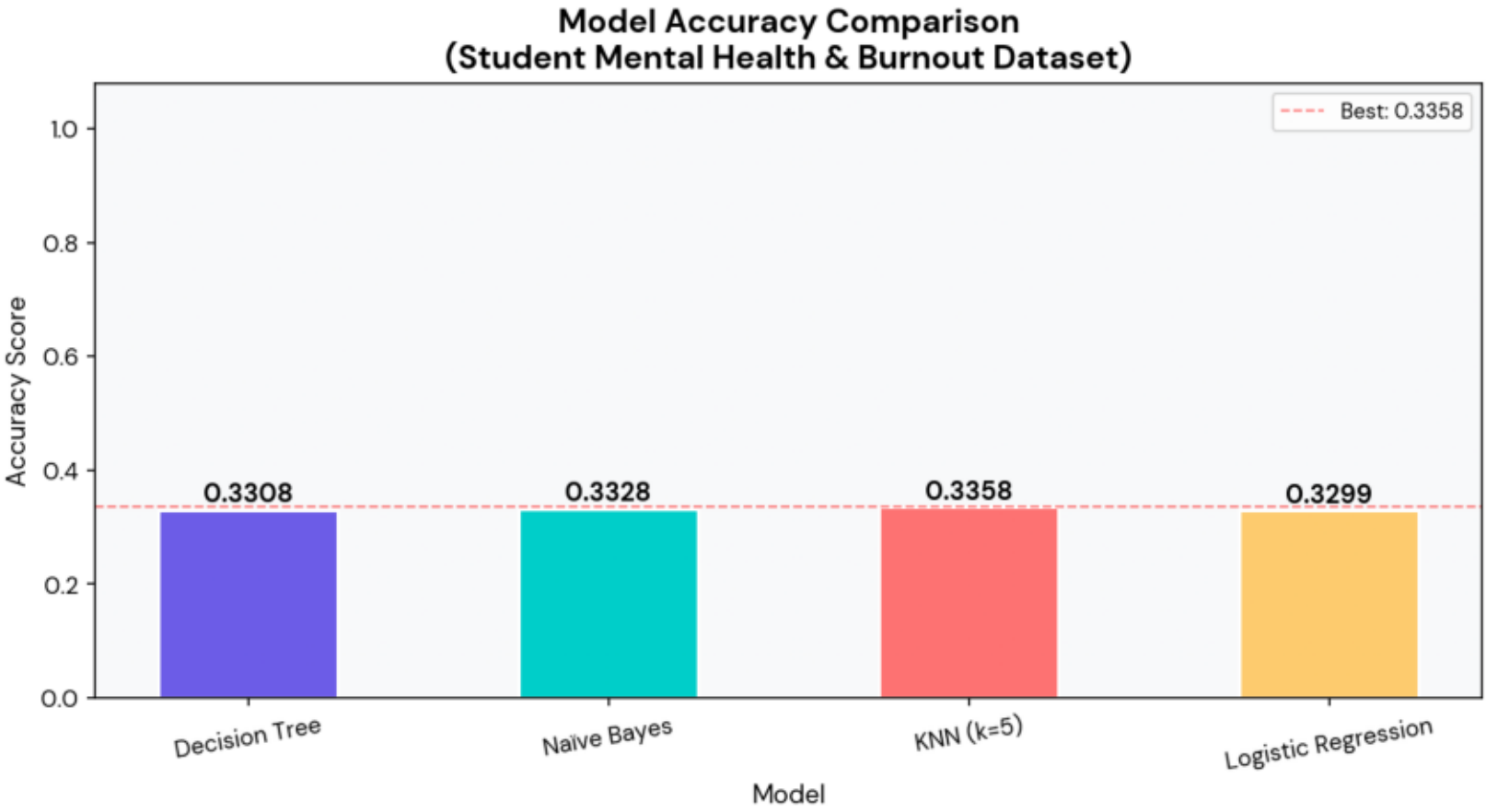

Chart 6 – Model Accuracy Comparison (Bar Chart)

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This bar chart provides a side-by-side comparison of the test-set accuracy achieved by each of the four trained models. Accuracy is the proportion of correctly classified samples out of all test samples (30,000 records). The dashed red line marks the best-performing model. Accuracy is a valid primary metric given the roughly balanced class distribution.

```
results = {"Decision Tree": dt_score, "Naive Bayes": nb_score,
           "KNN (k=5)": knn_score, "Logistic Regression": lr_score}
bars = ax.bar(results.keys(), results.values(), color=model_colors, width=0.5)
ax.axhline(y=max(results.values()), color="red", linestyle="--")
```

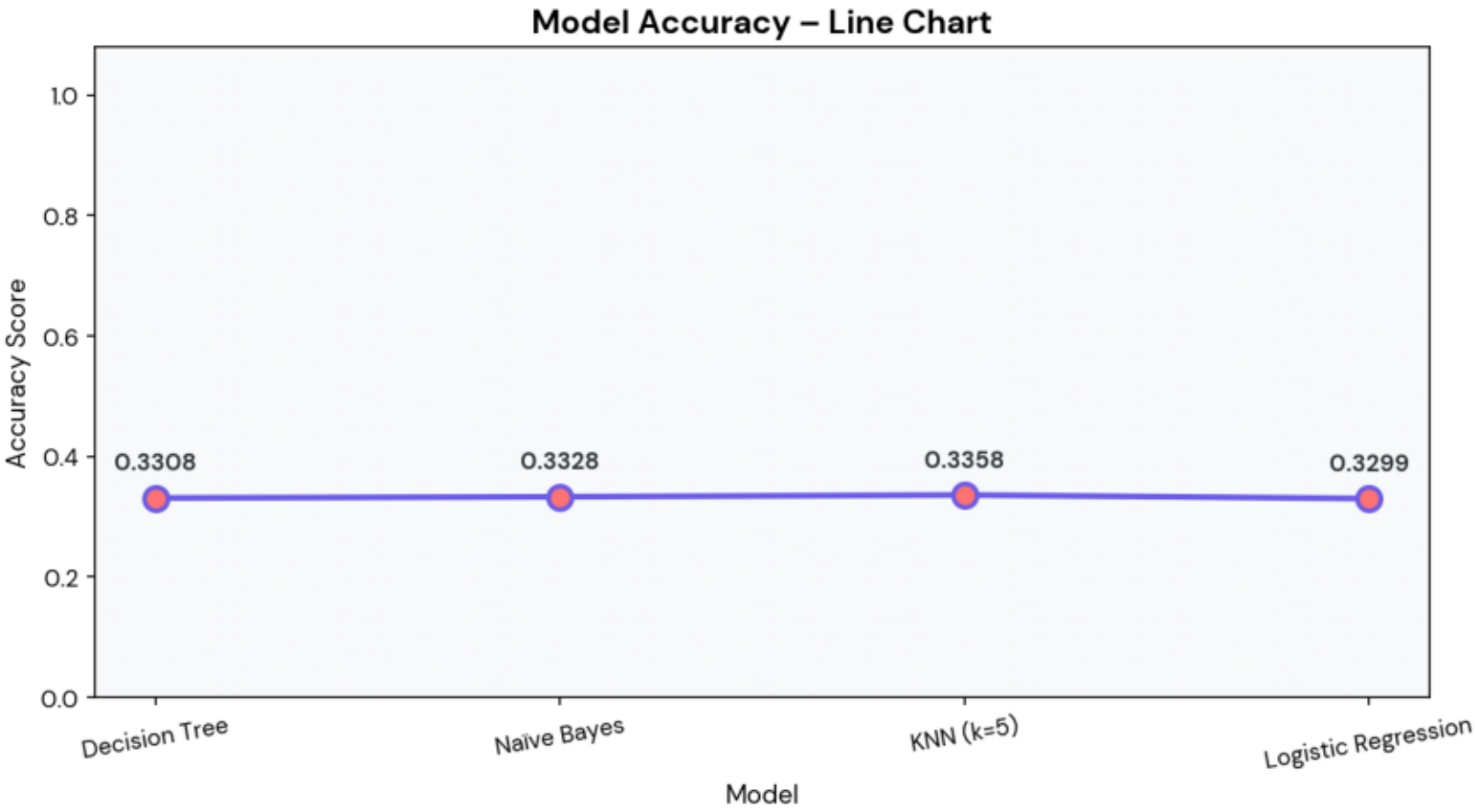

Chart 7 – Model Accuracy Comparison (Line Chart)

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This line chart presents the same accuracy values as Chart 6 but as a connected trend line, making relative differences between models easier to see at a glance. Each point is annotated with its exact accuracy score. A steeper slope between two adjacent models indicates a more significant performance difference.

```
ax.plot(list(results.keys()), list(results.values()),
        marker="o", markersize=10, linewidth=2.5, color="#6C5CE7")
for i, (name, acc) in enumerate(results.items()):
    ax.annotate(f"{acc:.4f}", xy=(i, acc), xytext=(0, 12),
                textcoords="offset points", ha="center")
```

Model Accuracy Summary

The table below summarises the test-set accuracy of all four trained models. Accuracy is defined as the proportion of correctly classified samples out of the total test samples (30,000 records — 20% of the full dataset).

Decision Tree	: 0.3308	(33.08%)
Naïve Bayes	: 0.3328	(33.28%)
KNN (k=5)	: 0.3358	(33.58%)
Logistic Regression	: 0.3299	(32.99%)

Best performing model: KNN (k=5) (33.58%)